Final Project Proposal

Sales Forecasting and Profitability Analysis System

Contributed Students

Student Name	Role	Contribution
[Khaled AbdelRahman **]**	**Project Leader**	Coordinated all milestones, managed timeline, and oversaw documentation & submission.
[Rahma Saber**]**	**Data Analyst**	Collected and cleaned dataset, performed EDA and statistical analysis.
[Ahmed Osama**]**	**Machine Learning Engineer**	Developed Prophet and Naïve forecasting models and validated predictions.
[Malak Shreif **]**	**UI/UX & Visualization Designer**	Created interactive dashboards and designed data visualizations for insights presentation.

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1. Project Planning

1.1 Project Overview

This project aims to develop a **Sales Forecasting and Profitability Analysis System** for the automotive retail sector.
By analyzing historical sales data, exploring key patterns, and applying predictive models, the system supports better **business decision-making** and **inventory management**.

1.2 Objectives

- Collect, clean, and preprocess historical car sales data.
- Identify trends, seasonality, and external factors influencing sales and profit.
- Develop machine learning–based forecasting models (Prophet).
- Provide actionable insights through visual dashboards and data-driven recommendations.

1.3 Project Scope

The project focuses on:

- **Sales and Profit Trends** (2018–2024)
- **Brand and Model Performance**
- **Seasonality Analysis** (monthly and quarterly patterns)
- **Weekend vs. Weekday sales**
- **Used vs. New Car Profitability**
- **Sales and Profit Forecasts for 2025–2026**

1.4 Tools and Technologies

Category	Tools / Libraries
Data Collection & Cleaning	Pandas, NumPy
Analysis & Visualization	Matplotlib, Seaborn, Plotly
Forecasting Models	Facebook Prophet, Naïve Forecasting
Dashboard / UI	Flask Api
Documentation & Reporting	Jupyter Notebook, MS Word / PDF

1.5 Timeline

Milestone	Description	Status
1	Data Collection, Exploration & Preprocessing	⍰ Completed
2	Advanced Data Analysis & Feature Engineering	⍰ Completed
3	Forecasting Model Development	⍰ Completed
4	Dashboard & Documentation	⍰ In Progress

2. Stakeholder Analysis

2.1 Stakeholder Identification

Stakeholder	Description	Role in Project
Dealership Management	Business decision-makers and executives.	Use forecasts to plan strategies, budgets, and promotions.
Sales & Marketing Teams	Staff responsible for sales operations and campaigns.	Use insights to optimize promotions, timing, and pricing.
Inventory Managers	Manage car stock and procurement.	Adjust inventory levels based on forecasted demand.
Financial Department	Handles budgeting, profit monitoring, and pricing.	Uses profitability insights to plan cost control strategies.
Customers	End-users indirectly affected by optimized pricing and availability.	Benefit from better deals, promotions, and stock management.

2.2 Stakeholder Interests and Impact

Stakeholder	Main Interest	Expected Impact
Dealership Management	Improved strategic decisions	Increased profit margins
Sales Teams	Accurate sales forecasting	Better marketing campaign results
Inventory Managers	Demand prediction	Reduced overstock or shortages
Financial Teams	Profit optimization	More stable cash flow
Customers	Availability and fair pricing	Enhanced satisfaction and loyalty

3. Database

3.1 Data Sources

The dataset includes **historical car sales and profit data** from 2018 to 2025, containing:

- Date (daily/weekly)
- Car Make and Model
- Units Sold
- Sales Amount
- Profit
- Promotions / Discounts
- Holidays
- Car Age

Resource: **https://drive.google.com/drive/folders/1d4BdpBJc4R3sO_dEydy_GQjt7igOJxEV?usp=sharing**

4. UI/UX Design

4.1 System Overview

The system provides **interactive dashboards** allowing users to explore sales trends, profitability, and forecasts by brand, model, and time period.

4.2 User Interface Features

- **Homepage Dashboard:** KPIs for total sales, profit, and top brands.
- **Trend Visualization:** Time-series charts for sales and profit.
- **Seasonality View:** Monthly/seasonal comparison.
- **Brand Comparison:** Toyota, BMW, Audi, Mercedes, etc.
- **Forecasting Section:** Prophet & Naïve models with confidence intervals.

4.3 User Experience (UX) Goals

- **Clarity:** Simplify data interpretation.
- **Interactivity:** Add filters for brand/model/time.
- **Decision Support:** Annotate charts with key recommendations.

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5. Summary of Analytical Findings

5.1 Key Insights

Aspect	Findings
Sales vs Profit	Sales have grown rapidly from 2018–2024, but profit remains flat due to higher costs and discounting.
Brand Performance	Toyota leads in volume but luxury brands (BMW, Audi, Mercedes) lead in margin.
Model Popularity	Affordable models outsell luxury ones but show higher volatility.
Seasonality	Spring is the strongest season; Winter and Fall are weakest.
Weekday Effect	Weekday performance exceeds weekends significantly.
Car Age	Used cars (3–6 years) generate higher margins than new ones.

5.2 Forecasting Results

- **Sales Forecast (Prophet):** Upward trend expected for 2026–2027.

6. Recommendations

- Conduct a **cost and profitability breakdown** to address margin stagnation.
- Focus on **high-margin vehicles** and optimize the product mix.
- Apply **value-based pricing** or tiered models.
- Increase **marketing and inventory** during Spring; use **discounts** in off-peak seasons.
- Expand **used car segment** to maximize profitability.

7. Deliverables

Deliverable	Description
Data Exploration Report	Summary of trends, seasonality, and data quality
EDA Notebook	Visualizations and correlation analysis
Cleaned Dataset	Preprocessed and feature-engineered data
Data Analysis Report	Statistical insights and conclusions
Forecasting Models	Prophet & Naïve predictions
Dashboard Prototype	Interactive visualization
Final Documentation	Full project proposal and analysis report

8. References / Appendix

- Jupyter Notebook (EDA & Forecasting)

https://github.com/kira924/Sales-Forecasting-and-Optimization/blob/main/Milestone_2/Copy_of_Final_Project_sec_milestone.ipyn

- Dataset Source (Historical Sales Data 2018–2024)

https://drive.google.com/drive/folders/1d4BdpBJc4R3sO_dEydy_GQjt7igOJxEV?usp=sharing

- Prophet Forecast Outputs
https://github.com/kira924/Sales-Forecasting-and-Optimization/blob/main/Baseline_profit_prediction_model.pkl